

Yizhe Zhang

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Research Interest

My recent research focus on pushing the LLM to gain more *intuition*, especially on *code domain*, regarding: **1)** code LLM/agent; **2)** look-ahead planning; **3)** non-autoregressive text generations; **4)** latent/implicit reasoning **5)** coding-based AI scientist.

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Experience

05/2022 – present

Staff Research Scientist | Apple, Cupertino, CA

Used code domain as a testbed for planning and reasoning. Most recent work include:

[DiffuCoder](#) – diffusion-based code generation. [SWE-Agent](#) – autonomous code agents.

09/2021 – 05/2022

Research Scientist | Facebook AI, Menlo Park, CA

Worked on RAG and Summarization.

02/2018 – 09/2021

Senior Researcher | Microsoft Research, Redmond, WA

Worked on generative dialogue models

[DialoGPT](#) – Open-sourced pretrained chat model.

Education

08/2013 – 02/2018

Ph.D. in Machine Learning | Duke University, Durham, NC

08/2015 – 02/2018

M.Sc. in Statistics | Duke University, Durham, NC

08/2007 – 06/2011

B.Sc. in Physics | Nanjing University, Nanjing, China

Publications

Selected preprint

- Amin Karimi Monsefi, Nikhil Bhendawade, Manuel Rafael Ciosici, Dominic Culver, **Yizhe Zhang**, Irina Belousova. FS-DFM: Fast and Accurate Long Text Generation with Few-Step Diffusion Language Models. *arXiv (2025)*
- Shansan Gong, Ruixiang Zhang, Huangjie Zheng, Jiatao Gu, Navdeep Jaitly, Lingpeng Kong, **Yizhe Zhang**. DiffuCoder: Understanding and Improving Masked Diffusion Models for Code Generation. *arXiv (2025)*
- Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, Junteng Liu, Yuntian Deng, **Yizhe Zhang**, Junxian He. Learn to Reason Efficiently with Adaptive Length-based Reward Shaping. *arXiv (2025)*
- Deepto Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, **Yizhe Zhang**, Tom Rainforth. BED-LLM: Intelligent Information Gathering with LLMs and Bayesian Experimental Design. *arXiv (2025)*
- **Yizhe Zhang**, Richard Bai, Zijin Gu, Ruixiang Zhang, Jiatao Gu, Emmanuel Abbe, Samy Bengio, Navdeep Jaitly. What makes the preferred thinking direction for LLM in Multi-choice Questions? *arXiv (2025)*.
- Xiaogeng Liu, Zhiyuan Yu, **Yizhe Zhang**, Ning Zhang, Chaowei Xiao. Automatic and Universal Prompt Injection Attacks Against Large Language Models. *arXiv (2024)*.

Peer-reviewed Conferences and Journals (* equally contributed)

- Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, **Yizhe Zhang**, Huangjie Zheng, Tianrong Chen, Miguel Angel Bautista, Josh Susskind, Navdeep Jaitly. Flexible Language Modeling in Continuous Space with Transformer-based Autoregressive Flows. *NeurIPS (2025)*
- Jiayi Pan, Xingyao Wang, Graham Neubig, Navdeep Jaitly, Heng Ji, Alane Suhr, **Yizhe Zhang**. Training Software Engineering Agents and Verifiers with SWE-Gym. *ICML (2025)*
- Ruixiang Zhang, Shuangfei Zhai, **Yizhe Zhang**, James Thornton, Zijing Ou, Joshua Susskind, Navdeep Jaitly. Target Concrete Score Matching: A Holistic Framework for Discrete Diffusion. *ICML (2025)*
- Yifu Qiu, Varun Embar, **Yizhe Zhang**, Navdeep Jaitly, Shay B. Cohen, Benjamin Han. Eliciting In-context Retrieval and Reasoning for Long-context Large Language Models. *ACL findings (2025)*
- Shansan Gong, Shivam Agarwal, **Yizhe Zhang**, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei Bi, Jiawei Han, et al. Scaling Diffusion Language Models via Adaptation from Autoregressive Models. *ICLR (2024)*.
- Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. All-hands: An Open Platform for AI Software Developers as Generalist Agents. *ICLR (2024)*.
- Jiatao Gu, Yuyang Wang, **Yizhe Zhang**, Qihang Zhang, Dinghuai Zhang, Navdeep Jaitly, Josh Susskind, Shuangfei Zhai. DART: Denoising Autoregressive Transformer for Scalable Text-to-Image Generation. *ICLR (2024)*.

- Guoli Yin, Haoping Bai, Shuang Ma, Feng Nan, Yanchao Sun, Zhaoyang Xu, Shen Ma, Jiarui Lu, Xiang Kong, Anan Zhang, et al. MMAU: A Holistic Benchmark of Agent Capabilities Across Diverse Domains. *CoRR* (2024).
- Jiarui Lu, Thomas Holleis, **Yizhe Zhang**, Bernhard Aumayer, Feng Nan, Felix Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, et al. Toolsandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities. *NAACL* (2024).
- Jiatao Gu*, Ying Shen*, Shuangfei Zhai, **Yizhe Zhang**, Navdeep Jaitly, Joshua M. Susskind. Kaleido Diffusion: Improving Conditional Diffusion Models with Autoregressive Latent Modeling. *NeurIPS* (2024).
- Yong Lin, Skyler Seto, Maartje Ter Hoeve, Katherine Metcalf, Barry-John Theobald, Xuan Wang, **Yizhe Zhang**, Chen Huang, Tong Zhang. On the Limited Generalization Capability of the Implicit Reward Model Induced by Direct Preference Optimization. *EMNLP findings* (2024).
- Zhuofeng Wu, He Bai, Anan Zhang, Jiatao Gu, VG Vydiswaran, Navdeep Jaitly, **Yizhe Zhang**. Divide-or-Conquer? Which Part Should You Distill Your LLM? *EMNLP findings* (2024).
- **Yizhe Zhang***, He Bai*, Ruixiang Zhang*, Jiatao Gu, Shuangfei Zhai, Josh Susskind, Navdeep Jaitly. How Far Are We from Intelligent Visual Deductive Reasoning? *COLM* (2024)
- **Yizhe Zhang**, Jiarui Lu, Navdeep Jaitly. The Entity-Deduction Arena: A playground for probing the conversational reasoning and planning capabilities of LLMs. *ACL* (2024)
- Pratyush Maini, Skyler Seto, He Bai, David Grangier, **Yizhe Zhang**, Navdeep Jaitly. Rephrasing the web: A recipe for compute and data-efficient language modeling. *ACL* (2024)
- Dinghuai Zhang, **Yizhe Zhang**, Jiatao Gu, Ruixiang Zhang, Josh Susskind, Navdeep Jaitly, Shuangfei Zhai. Improving GFlowNets for Text-to-Image Diffusion Alignment. *ICML SPIGM workshop* (2024).
- Xingyao Wang, Yangyi Chen, Lifan Yuan, **Yizhe Zhang**, Yunzhu Li, Hao Peng, Heng Ji. Executable code actions elicit better llm agents. *ICML* (2024)
- Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Lingjie Liu, Joshua M Susskind. Boot: Data-free distillation of denoising diffusion models with bootstrapping. *ICML* (2024)
- Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Joshua M Susskind, Navdeep Jaitly. Matryoshka diffusion models. *ICLR* (2023)
- **Yizhe Zhang**, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Joshua Susskind, Navdeep Jaitly. PLANNER: generating diversified paragraph via latent language diffusion model. *NeurIPS* (2023)
- Shuangfei Zhai, Tatiana Likhomanenko, Etai Littwin, Dan Busbridge, Jason Ramapuram, **Yizhe Zhang**, Jiatao Gu, Joshua M Susskind. Stabilizing transformer training by preventing attention entropy collapse. *ICML* (2023)
- Felix Faltings, Michel Galley, Baolin Peng, Kianté Brantley, Weixin Cai, **Yizhe Zhang**, Jianfeng Gao, Bill Dolan. Interactive text generation. *EMNLP* (2023)

- Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Miguel Angel Bautista, Josh Susskind. f-dm: A multi-stage diffusion model via progressive signal transformation. *ICLR* (2023)
- Gyuwan Kim, Jinhyuk Lee, Barlas Oguz, Wenhan Xiong, **Yizhe Zhang**, Yashar Mehdad, William Yang Wang. Bridging the training-inference gap for dense phrase retrieval. *EMNLP Findings* (2022)
- **Yizhe Zhang***, Deng Cai*. Linearizing transformer with key-value memory. *EMNLP* (2022)
- Zhisong Zhang, **Yizhe Zhang**, Bill Dolan. Towards More Efficient Insertion Transformer with Fractional Positional Encoding. *EACL* (2021)
- Jiachang Liu, Dinghan Shen, **Yizhe Zhang**, Bill Dolan, Lawrence Carin, Weizhu Chen. What Makes Good In-Context Examples for GPT-3? *ACL DEELIO workshop* (2021)
- Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Bill Dolan. Automatic document sketching: Generating drafts from analogous texts. *ACL Findings* (2021)
- Tianyu Liu, **Yizhe Zhang**, Chris Brockett, Yi Mao, Zhifang Sui, Weizhu Chen, Bill Dolan. A token-level reference-free hallucination detection benchmark for free-form text generation. *ACL* (2022)
- **Yizhe Zhang**, Siqi Sun, Xiang Gao, Yuwei Fang, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Joint retrieval and generation training for grounded text generation. *AAAI* (2022)
- Jungo Kasai, Hao Peng, **Yizhe Zhang**, Dani Yogatama, Yi Mao, Weizhu Chen, Noah A Smith. Finetuning Pretrained Transformers into RNNs. *EMNLP (2021, Oral Presentation)*
- Dianqi Li, **Yizhe Zhang**, Hao Peng, Liqun Chen, Chris Brockett, Ming-Ting Sun, Bill Dolan. Contextualized perturbation for textual adversarial attack. *NAACL* (2021)
- Woon Sang Cho, **Yizhe Zhang**, Sudha Rao, Asli Celikyilmaz, Chenyan Xiong, Jianfeng Gao, Mengdi Wang, Bill Dolan. Unsupervised Common Question Generation from Multiple Documents using Reinforced Contrastive Coordinator. *EACL* (2021)
- Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Xiang Gao, Chris Quirk, Rik Koncel-Kedziorski, Jianfeng Gao, Hannaneh Hajishirzi, Mari Ostendorf, Bill Dolan. A Controllable Model of Grounded Response Generation. *AAAI* (2021)
- Ramakanth Pasunuru, Asli Celikyilmaz, Michel Galley, Chenyan Xiong, **Yizhe Zhang**, Mohit Bansal, Jianfeng Gao. Data Augmentation for Abstractive Query-Focused Multi-Document Summarization. *AAAI* (2021)
- **Yizhe Zhang**, Xiang Gao, Sungjin Lee, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Consistent Dialogue Generation with Self-supervised Feature Learning. *SIGDIAL* (2020).
- **Yizhe Zhang**, Guoyin Wang, Chunyuan Li, Zhe Gan, Chris Brockett, Bill Dolan. POINTER: Constrained Text Generation via Insertion-based Generative Pre-training. *EMNLP* (2020)
- Xiang Gao, **Yizhe Zhang**, Michel Galley, Chris Brockett and Bill Dolan. Dialogue Response Ranking Training with Large-Scale Human Feedback Data. *EMNLP* (2020)
- Chunyuan Li, Xiang Gao, Yuan Li, Xiujuan Li, Baolin Peng, **Yizhe Zhang**, Jianfeng Gao. Optimus: Organizing Sentences via Pre-trained Modeling of a Latent Space. *EMNLP* (2020)

- Jianqiao Li, Chunyuan Li, Guoyin Wang, Hao Fu, Yuhchen Lin, Liqun Chen, **Yizhe Zhang** and Lawrence Carin. Improving Text Generation with Student-Forcing Optimal Transport. *EMNLP (2020)*
- Yu Cheng, Zhe Gan, **Yizhe Zhang**, Oussama Elachqar, Dianqi Li, Jingjing Liu. Contextual Text Style Transfer. *Findings of EMNLP (2020)*
- Shuyang Dai, Yu Cheng, **Yizhe Zhang**, Zhe Gan, Jingjing Liu and Lawrence Carin. Contrastively Smoothed Class Alignment for Unsupervised Domain Adaptation. *ACCV (2020)*
- Xinnuo Xu, **Yizhe Zhang**, Lars Liden, Sungjin Lee. Datasets and Benchmarks for Task-Oriented Log Dialogue Ranking Task. *Interspeech (2020)*
- Siyang Yuan, Ke Bai, Liqun Chen, **Yizhe Zhang**, Chenyang Tao, Chunyuan Li, Guoyin Wang, Ricardo Henao, Lawrence Carin. Weakly supervised cross-domain alignment with optimal transport Task. *BMVC (2020)*
- **Yizhe Zhang**, Siqi Sun, Michel Galley, Yen-Chun Chen, Chris Brockett, Xiang Gao, Jianfeng Gao, Jingjing Liu, Bill Dolan. DialoGPT: Large-Scale Generative Pre-training for Conversational Response Generation. *demo track, ACL (2020)*
- Yichen Huang*, **Yizhe Zhang***, Oussama Elachqar, Yu Cheng. INSET: Sentence Infilling with Inter-sentential Generative Pre-training. *ACL (2020)*
- Pengyu Cheng, Renqiang Min, Dinghan Shen, Christopher Malon, **Yizhe Zhang**, Yitong Li and Lawrence Carin. Improving Disentangled Text Representation Learning with Information Theoretical Guidance. *ACL (2020)*
- Xinjie Fan, **Yizhe Zhang**, Zhendong Wang, Mingyuan Zhou. Adaptive Correlated Monte Carlo for Contextual Categorical Sequence Generation. *ICLR (2020)*
- Yuan Li, Chunyuan Li, **Yizhe Zhang**, Xiujun Li, Guoqing Zheng, Lawrence Carin, Jianfeng Gao. Complementary Auxiliary Classifiers for Label-Conditional Text Generation. *AAAI (2020)*
- Liqun Chen, Ke Bai, Chenyang Tao, **Yizhe Zhang**, Guoyin Wang, Wenlin Wang, Ricardo Henao, Lawrence Carin. Sequence Generation with Optimal-Transport-Enhanced Reinforcement Learning. *AAAI (2020)*
- Xiang Gao, **Yizhe Zhang**, Sungjin Lee, Michel Galley, Chris Brockett, Jianfeng Gao and Bill Dolan. Structuring latent spaces for stylized response generation. *EMNLP (2019)*
- Dianqi Li, **Yizhe Zhang**, Zhe Gan, Yu Cheng, Chris Brockett, Ming-Ting Sun and Bill Dolan. Domain Adaptive Text Style Transfer. *EMNLP (2019)*
- Xinnuo Xu, **Yizhe Zhang**, Lars Liden and Sungjin Lee. Unsupervised Dialogue Spectrum Generation for Log Dialogue Ranking. *SIGDIAL (2019), Best paper nomination*
- Liqun Chen, Guoyin Wang, Chenyang Tao, Dinghan Shen, **Yizhe Zhang** and Lawrence Carin. Improving Textual Network Embedding with Global Attention via Optimal Transport. *ACL (2019)*

- Dinghan Shen, Asli Celikyilmaz, **Yizhe Zhang**, Liqun Chen, Xin Wang, Jianfeng Gao, Lawrence Carin. Towards Generating Long and Coherent Text with Multi-Level Latent Variable Models. *ACL (2019)*
- Xiang Gao, Sungjin Lee, **Yizhe Zhang**, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Jointly Optimizing Diversity and Relevance in Neural Response Generation. *NAACL (2019)*
- Liqun Chen, **Yizhe Zhang**, Ruiyi Zhang, Chenyang Tao, Zhe Gan, Haichao Zhang, Bai Li, Dinghan Shen, Changyou Chen, Lawrence Carin. Improving Sequence-to-Sequence Learning via Optimal Transport. *ICLR (2019)*
- **Yizhe Zhang**, Michel Galley, Jianfeng Gao, Zhe Gan, Xiujun Li, Chris Brockett, Bill Dolan. Generating Informative and Diverse Conversational Responses via Adversarial Information Maximization. *NeurIPS (2018)*
- Liqun Chen, Shuyang Dai, Chenyang Tao, Dinghan Shen, Zhe Gan, Haichao Zhang, **Yizhe Zhang**, Lawrence Carin. Adversarial Text Generation via Feature-Mover's Distance. *NeurIPS (2018)*
- Yunchen Pu, Shuyang Dai, **Yizhe Zhang**, Zhe Gan and Lawrence Carin. Multi-Domain Joint Distribution Learning with Generative Adversarial Nets. *ICML (2018)*
- Dinghan Shen, Guoyin Wang, Wenlin Wang, Martin Renqiang Min, Qinliang Su, **Yizhe Zhang**, Chunyuan Li, Ricardo Henao and Lawrence Carin. On Simple Word-Embedding-Based Models and Associated Pooling Mechanisms. *ACL (2018)*
- Guoyin Wang, Chunyuan Li, Wenlin Wang, **Yizhe Zhang**, Dinghan Shen, Xinyuan Zhang, Ricardo Henao and Lawrence Carin. Joint Embedding of Words and Labels for Text Classification. *ACL (2018)*
- Dinghan Shen, **Yizhe Zhang**, Ricardo Henao, Qinliang Su, Lawrence Carin. Deconvolutional Latent-Variable Model for Text Sequence Matching. *AAAI (2018)*.
- Wenlin Wang, Piyush Rai, Yunchen Pu, Kai Fan, **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. A Flexible Probabilistic Framework for Learning to Predict Unseen Classes. *AAAI (2018)*.
- Zhe Gan, Liqun Chen, Weiyao Wang, Yunchen Pu, **Yizhe Zhang**, Lawrence Carin. Triangle Generative Adversarial Networks. *NIPS (2017)*.
- **Yizhe Zhang**, Changyou Chen, Zhe Gan, Lawrence Carin. Stochastic Gradient Monomial Gamma Sampler. *ICML (2017)*.
- **Yizhe Zhang**, Zhe Gan, Zhi Chen, Lawrence Carin. Adversarial Feature Matching for Text Generation. *ICML (2017)*.
- **Yizhe Zhang**, Xiangyu Wang, Changyou Chen, Lawrence Carin. Towards Unifying Hamiltonian Monte Carlo and Slice Sampling. *NIPS (2016)*.
- Changyou Chen, Nan Ding, Chunyuan Li, **Yizhe Zhang**, Lawrence Carin. Distributed Bayesian Learning with Stochastic Gradient MCMC. *NIPS (2016)*.
- **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. Dynamic Poisson Factor Analysis. *ICDM (2016)*.

- Kai Fan, **Yizhe Zhang**, Katherine Heller. Triply Stochastic Variational Inference for Non-linear Beta Process Factor Analysis. *ICDM (2016)*.
- **Yizhe Zhang**, Ricardo Henao, Jianling Zhong, Lawrence Carin, Alexander Hartemink. Learning a Hybrid Architecture for Sequence Regression and Annotation. *AAAI (2016)*.
- **Yizhe Zhang**, Ricardo Henao, Chunyuan Li, Lawrence Carin. Bayesian Dictionary Learning with Gaussian Processes and Sigmoid Belief Networks. *IJCAI (2016)*.
- **Yizhe Zhang**, Changyou Chen, Ricardo Henao, Lawrence Carin. Laplacian Hamiltonian Monte Carlo. *ECML (2016)*.
- **Yizhe Zhang**, Yupeng He and Chaochun Wei. MOST+: a Motif Finding Approach Combining Genomic Sequence and Heterogeneous Genome-wide Signatures. *BMC Genomics (2015)*.
- Yupeng He, **Yizhe Zhang**, Guangyong Zheng and Chaochun Wei. CRF-based Transcription Factor Binding Site Finding System. *BMC Genomics (2012)*.
- Jiemeng Liu, Haifeng Wang, Hongxing Yang, **Yizhe Zhang**, Jinfeng Wang, Fangqing Zhao and Ji Qi. Composition-based Classification of Short Metagenomic Sequences Elucidates the Landscapes of Taxonomic and Functional Enrichment of Microorganisms. *Nucleic Acids Research (2012)*

Workshop, Demo, Miscellaneous

- **Yizhe Zhang**, Navdeep Jaitly. SAGE: Steering Dialog Generation with Future-Aware State-Action Augmentation. *EMNLP Workshop on NLP Perspectives (2025)*.
- Georgia Gabriela Sampaio, Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, Josh Susskind, Navdeep Jaitly, **Yizhe Zhang**. TypeScore: A Text Fidelity Metric for Text-to-Image Generative Models. *arXiv (2024)*.
- Ying Shen, **Yizhe Zhang**, Shuangfei Zhai, Lifu Huang, Joshua M. Susskind, Jiatao Gu. Many-to-Many Image Generation with Auto-Regressive Diffusion Models. *ICML 2024 Workshop on Structured Probabilistic Inference Generative Modeling (2024)*
- Tatiana Likhomanenko, Luke Carlson, Richard He Bai, Zijin Gu, Han Tran, Zakaria Aldeneh, **Yizhe Zhang**, Ruixiang Zhang, Huangjie Zheng, Navdeep Jaitly. ChipChat: Low-Latency Cascaded Conversational Agent in MLX. *ASRU demo track (2025)*

Teaching

- Advanced Machine Learning @Duke (STA571). Instructor: *Katherine Heller*
- Probabilistic Machine Learning @Duke (CS571). Instructor: *Cynthia Rudin*

Rewards

- **Stanford top 2% scientists (since 2023)**
- NeurIPS top 5% reviewer award. (2018)
- Department Fellowship. (2008-2011)
- National Excellent Graduate Scholarship (top 1%). (2012)
- Travel award: NIPS (2015, 2016), ICML (2017), ICDM (2016), IJCAI (2016), AAAI (2016)

Professional Services

Area Chair: NeurIPS (since 2020), ICML (since 2022), ICLR (since 2023), ACL (2020-2021), EMNLP (2022), NAACL (2023) and AAAI (2018-2021)

Action Editor: TMLR (since 2023), ARR (since 2023)

Organization committee: ACL (2020)

Proficiency

- Pytorch, Tensorflow, C/C++, Python, Java, Lua, MATLAB and R.